



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 3 (Summary and Tutorial)

Topic: Integration Techniques

Summary for Integration Techniques

General Tips

- For indefinite integrals, do not forget the “+ C”
- Can always check your answer by differentiating / use GC for definite integrals
- Before using integration by parts when you see a product of two functions, ask yourself if the expression is of the form $f'(x)(f(x))^n$ or $\frac{f'(x)}{f(x)}$.

Overview of Integration Techniques

In general, you should know how to find the anti-derivatives of the following classes of functions:

- **L**ogarithmic ($\ln x$)
- **I**nverse trigo ($\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$)
- **A**lgebraic (polynomials, rational functions $\frac{P(x)}{Q(x)}$, where P and Q are polynomials)
- **T**rigonometric (6 basic trigonometric functions, and rational functions of them)
- **E**xponential (e^x)

This is also the ‘order’ for integration by parts (except for the rational functions); with the functions at the end of the list the functions you would usually choose to integrate first when doing integration by parts.

Before we talk about integration by parts, let us review the integration techniques involving the **E**xponential, **T**rigonometric and **A**lgebraic functions first.

Exponential functions

Nothing much to say here, you should know how to find the anti-derivatives of such functions.

$\int e^{ax} dx = \frac{1}{a} e^{ax} + C$ is your staple here.

Trigonometric functions

You need to know the anti-derivatives of the 6 basic trigo functions (sin and cos obviously, the other 4 are in MF27). For higher powers of the trigo functions and product of several of them, we have to use trigonometric identities to help us.

In general, the idea is to **linearize products** (such as by using the double angle formulae or trigo identities)

- $\sin^2 x, \cos^2 x$ can be reduced to $\cos 2x$ (can be done repeatedly)
- $\tan^2 x, \cot^2 x$ can be expressed in terms of $\sec^2 x, \operatorname{cosec}^2 x$ using the trigonometric identities $(1 + \tan^2 x = \sec^2 x, 1 + \cot^2 x = \operatorname{cosec}^2 x)$. Their respective antiderivatives are $\tan x$ and $-\cot x$.

Basic trigonometric functions (highlighted are in MF 27)

$f(x)$	$\int f(x) dx$
$\sin x$	$-\cos x$
$\cos x$	$\sin x$
$\tan x$	$-\ln \cos x = \ln \sec x $
$\sec x$	$\ln \sec x + \tan x $
$\operatorname{cosec} x$	$-\ln \operatorname{cosec} x + \cot x $
$\cot x$	$\ln \sin x $

$f(x)$	$\int f(x) dx$
$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$	$\frac{1}{2}\left(x - \frac{1}{2}\sin 2x\right)$
$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$	$\frac{1}{2}\left(x + \frac{1}{2}\sin 2x\right)$
$\tan^2 x = \sec^2 x - 1$	$\tan x - x$
$\sec^2 x$	$\tan x$
$\operatorname{cosec}^2 x$	$-\cot x$
$\cot^2 x = \operatorname{cosec}^2 x - 1$	$-\cot x - x$

- Be aware that on the 3rd page of MF27, the following **derivatives** are given:

$f(x)$	$f'(x)$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\sec x$	$\sec x \tan x$

So, it means that

$$1) \int -\operatorname{cosec} x \cot x dx = \operatorname{cosec} x + C. \text{ That is, } \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + D$$

$$2) \int \sec x \tan x dx = \sec x + C$$

- $f'(x)(f(x))^n$ or $\frac{f'(x)}{f(x)}$.

Some integrals like $\int \frac{\sin x}{2 + \cos x} dx$ are of the form $\int \frac{f'(x)}{f(x)} dx$. Some like

$\int \sin x \sec^5 x dx$ are of the form $\int f'(x)(f(x))^n dx$. Try to 'spot' these forms first before resorting to brute force (converting everything to sine and cosine and using double angle formula)! In these questions, it may be helpful to write down the $f'(x)$ first (it is usually sine or cosine, and in general would be the derivative of a trigonometric function).

Algebraic functions

You should know, $\int x^n dx = \begin{cases} \frac{1}{n+1} x^{n+1} + C, & \text{if } n \neq -1 \\ \ln|x| + C, & \text{if } n = -1 \end{cases}$.

This section is dedicated to finding the anti-derivatives for functions of the form $\frac{P(x)}{Q(x)}$.

- Long division (if the fraction is improper, i.e. $\deg P \geq \deg Q$)
- Factorize $Q(x)$ into a product of linear and quadratic factors
- Partial fractions based on the linear/quadratic factors
- There will be 3 types of terms $\frac{1}{x-a}$, $\frac{1}{(x-a)^2}$ and $\frac{Ax+B}{x^2+a^2}$. The anti-derivatives are respectively of the form $\ln|x-a|$, $-\frac{1}{x-a}$ and $\frac{A}{2} \ln(x^2+a^2) + \frac{B}{a} \tan^{-1}\left(\frac{x}{a}\right)$.

Example

$$\begin{aligned} \int \frac{1}{1+x^3} dx &= \int \frac{1}{(1+x)(x^2-x+1)} dx \quad (\text{factorize denominator into linear and quadratic factors}) \\ &= \int \frac{\frac{1}{3}}{(1+x)} + \frac{-\frac{x}{3} + \frac{2}{3}}{(x^2-x+1)} dx \quad (\text{partial fractions}) \\ &= \frac{1}{3} \ln|1+x| - \frac{1}{3} \int \frac{1}{2} \frac{(2x-1)-3}{x^2-x+1} dx \quad (\text{split into the } \ln \text{ and } \tan^{-1} \text{ terms}) \\ &= \frac{1}{3} \ln|1+x| - \frac{1}{6} \ln|x^2-x+1| + \frac{1}{2} \int \frac{1}{\left(x-\frac{1}{2}\right)^2 + \frac{3}{4}} dx \quad (\text{complete the square for the } \tan^{-1} \text{ term}) \\ &= \frac{1}{3} \ln|1+x| - \frac{1}{6} \ln(x^2-x+1) + \frac{1}{\sqrt{3}} \tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right) dx + C \quad (\text{since } x^2-x+1 = \left(x-\frac{1}{2}\right)^2 + \frac{3}{4} > 0 \text{ for all } x \in \mathbb{R}) \end{aligned}$$

There is another algebraic form $\int \frac{1}{\sqrt{a^2-x^2}} dx$ which you will need to know how to evaluate, and it is also given in the **MF 27**,

$f(x)$	$\int f(x) dx$
$\frac{1}{x^2+a^2}$	$\frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$
$\frac{1}{\sqrt{a^2-x^2}}$	$\sin^{-1}\left(\frac{x}{a}\right)$
$\frac{1}{x^2-a^2}$	$\frac{1}{2a} \ln\left(\frac{x-a}{x+a}\right)$
$\frac{1}{a^2-x^2}$	$\frac{1}{2a} \ln\left(\frac{a+x}{a-x}\right)$

Before we discuss how to find the antiderivatives of **L**ogarithmic and **I**nverse Trigonometric functions, we need to talk about Integration by Parts.

Integration by Parts

This is a useful tool when dealing with anti-derivatives of products of functions (as it is essentially the reverse of the product rule).

$$\int u \frac{dv}{dx} dx = \underbrace{v}_{\text{Integrate}} \underbrace{u}_{\text{Keep}} - \int \underbrace{v}_{\text{Integrate}} \underbrace{\frac{du}{dx}}_{\text{Differentiate}} dx$$

$$\int_a^b u \frac{dv}{dx} dx = [v(x)u(x)]_a^b - \int_a^b v \frac{du}{dx} dx$$

When we have a combination of 2 or more (usually 2 for H2) different types of functions, the LIATE rule can help us determine which function we should integrate first. The following is non-exhaustive, but shows typical combinations asked at the A-levels:

Logarithmic + Algebraic, Inverse Trig + Algebraic

$\int P(x) \ln x dx, \int P(x) \tan^{-1} x dx$. Clearly the only choice here is to differentiate the logarithmic or inverse trigonometric function and integrate the polynomial.

Example:

$$\int \underbrace{x^2}_{\frac{dv}{dx}} \underbrace{\ln x}_u dx = \underbrace{\frac{x^3}{3}}_{\text{Integrate}} \underbrace{\ln x}_{\text{Keep}} - \int \underbrace{\frac{x^3}{3}}_{\text{Integrate}} \underbrace{\left(\frac{1}{x}\right)}_{\text{Differentiate}} dx$$

$$= \frac{x^3 \ln x}{3} - \frac{x^3}{9} + C$$

$$\int \underbrace{x}_{\frac{dv}{dx}} \underbrace{\tan^{-1} x}_u dx = \underbrace{\frac{x^2}{2}}_{\text{Integrate}} \underbrace{\tan^{-1} x}_{\text{Keep}} - \int \underbrace{\frac{x^2}{2}}_{\text{Integrate}} \underbrace{\left(\frac{1}{1+x^2}\right)}_{\text{Differentiate}} dx$$

$$= \frac{x^2 \tan^{-1} x}{2} - \frac{1}{2} \int 1 - \frac{1}{1+x^2} dx \quad (\text{long division since improper rational function})$$

$$= \frac{x^2 \tan^{-1} x}{2} - \frac{1}{2} (x - \tan^{-1} x) + C$$

Algebraic + Trigonometric, Algebraic + Exponential

$\int P(x) \cos x \, dx, \int P(x) e^x \, dx$, where P is a polynomial. We repeatedly integrate the trigonometric/exponential function, and differentiate the polynomial until the degree of the polynomial reaches 0.

Example

$$\begin{aligned} \int \underbrace{x^2}_u \underbrace{\cos x}_{\frac{dv}{dx}} \, dx &= \underbrace{(\sin x)}_{\text{Integrate}} \underbrace{x^2}_{\text{Keep}} - \int \underbrace{(\sin x)}_{\text{Integrate}} \underbrace{(2x)}_{\text{Differentiate}} \, dx \\ &= x^2 \sin x - 2 \int x \sin x \, dx \\ &= x^2 \sin x - 2 \left(-x \cos x - \int (-\cos x) \, dx \right) \\ &= x^2 \sin x + 2x \cos x - 2 \sin x + C \end{aligned}$$

It is highly advisable that you use () and also factorize/multiply coefficients (especially those negative signs) with great prudence.

Trigonometric + Exponential $\int e^{ax} \cos bx \, dx, \int e^{ax} \sin bx \, dx$

Integration by parts 2 times (you can choose to **either integrate the exponential or trigonometric function**, but it must be the same function that you integrate 2 times)

Example

$$\begin{aligned} \int e^{2x} \cos 3x \, dx &= \frac{1}{2} e^{2x} \cos 3x - \int \frac{1}{2} e^{2x} (-3 \sin 3x) \, dx \\ &= \frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} \int e^{2x} \sin 3x \, dx \\ &= \frac{1}{2} e^{2x} \cos 3x + \frac{3}{2} \left(\frac{1}{2} e^{2x} \sin 3x - \int \frac{1}{2} e^{2x} (3 \cos 3x) \, dx \right) \\ &= \frac{1}{2} e^{2x} \cos 3x + \frac{3}{4} e^{2x} \sin 3x - \frac{9}{4} \int e^{2x} \cos 3x \, dx \end{aligned}$$

Now rearranging all the terms involving the integral to the LHS, we will have

$$\begin{aligned} \frac{13}{4} \int e^{2x} \cos 3x \, dx &= \frac{1}{2} e^{2x} \cos 3x + \frac{3}{4} e^{2x} \sin 3x + C' \\ \Rightarrow \int e^{2x} \cos 3x \, dx &= \frac{2}{13} e^{2x} \cos 3x + \frac{3}{13} e^{2x} \sin 3x + C \end{aligned}$$

Now let us discuss the techniques that allow us to find the antiderivatives of Logarithmic and Inverse Trigonometric functions. In essence, it is Integration by Parts, using a 'trick' that you need to be familiar with. When faced with finding the anti-derivative of a logarithmic or inverse trigonometric function by itself, we often write $\int f(x) \, dx = \int 1 \cdot f(x) \, dx$ so as to allow us to integrate 1 and differentiate the logarithmic or inverse trigonometric function when using integration by parts.

Logarithmic functions

These are typically done by writing $\int f(x) \, dx = \int \underbrace{\frac{1}{\frac{dv}{dx}}}_{u} \cdot \underbrace{f(x)}_u \, dx = xf(x) - \int xf'(x) \, dx$.

Example: $\int \ln x \, dx = \int \underbrace{\frac{1}{\frac{dv}{dx}}}_{\frac{1}{x}} \cdot \underbrace{\ln x}_u \, dx = x \ln|x| - \int x \left(\frac{1}{x}\right) \, dx = x \ln|x| - x + C$.

Inverse trigonometric functions

Same as above (for the logarithmic functions).

Example: $\int \tan^{-1} x \, dx = \int \underbrace{\frac{1}{\frac{dv}{dx}}}_{\frac{1}{1+x^2}} \cdot \underbrace{\tan^{-1} x}_u \, dx = x \tan^{-1} x - \int \frac{x}{1+x^2} \, dx = x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) + C$.

Integration by Substitution

When a question requires you to do a substitution, the substitution will be provided for you. You need to remember that there are generally 3 steps (for either indefinite/definite integrals).

Steps	Indefinite Integral
Question	Use the substitution $x = 2 \tan \theta$ to find $\int \frac{1}{\sqrt{x^2 + 4}} \, dx$.
Substitute Limit	<i>There are no upper/lower limits in an indefinite integral</i>
Substitute dx (diff. the relation given)	$x = 2 \tan \theta \Rightarrow \frac{dx}{d\theta} = 2 \sec^2 \theta$ Therefore, $dx = 2 \sec^2 \theta \, d\theta$
Substitute integrand	$\frac{1}{\sqrt{x^2 + 4}} = \frac{1}{\sqrt{4 \tan^2 \theta + 4}} = \frac{1}{2 \sec \theta}$
	$\int \frac{1}{\sqrt{x^2 + 4}} \, dx$ $= \int \frac{1}{2 \sec \theta} 2 \sec^2 \theta \, d\theta$ $= \int \sec \theta \, d\theta$ $= \ln \sec \theta + \tan \theta + C$
Substitute back to original variable	$= \ln \sec \theta + \tan \theta + C$ $= \ln \left \frac{\sqrt{x^2 + 4} + x}{2} \right + C$

Steps	Definite Integral
Question	Use the substitution $x = 2 \tan \theta$ to find $\int_0^2 \frac{1}{\sqrt{x^2 + 4}} \, dx$
Substitute Limit	Upper limit: $x = 2$ So $2 = 2 \tan \theta \Rightarrow \theta = \frac{\pi}{4}$ Lower limit: $x = 0$ So $0 = 2 \tan \theta \Rightarrow \theta = 0$
Substitute dx (diff. the relation given)	$x = 2 \tan \theta \Rightarrow \frac{dx}{d\theta} = 2 \sec^2 \theta$ Therefore, $dx = 2 \sec^2 \theta \, d\theta$
Substitute integrand	$\frac{1}{\sqrt{x^2 + 4}} = \frac{1}{\sqrt{4 \tan^2 \theta + 4}} = \frac{1}{2 \sec \theta}$
	$\int_0^2 \frac{1}{\sqrt{x^2 + 4}} \, dx$ $= \int_0^{\frac{\pi}{4}} \frac{1}{2 \sec \theta} 2 \sec^2 \theta \, d\theta$ $= \int_0^{\frac{\pi}{4}} \sec \theta \, d\theta$ $= [\ln \sec \theta + \tan \theta]_0^{\frac{\pi}{4}} = \ln(1 + \sqrt{2})$
Substitute back to original variable	<i>No need since definite integral gives a real value (if it exists)</i>

Revision Tutorial Questions

1. Evaluate the following integrals.

(a) $\int \frac{x^2 - 2}{x^2 - x - 2} dx$	(b) $\int \sin 2x \cos^4 2x dx$	(c) $\int \sin^3 x dx$
(d) $\int \frac{x^2 - x + 1}{x^2 + 9} dx$	(e) $\int \frac{x^2 - 2}{(x-1)^2(x^2 + 4)} dx$	(f) $\int \frac{1}{x \ln x} dx$
(g) $\int x^{\frac{3}{2}} \ln x dx$	(h) $\int x \tan^{-1} 2x dx$	(i) $\int (1 + \tan x)^2 dx$
(j) $\int \frac{x-1}{\sqrt{-x^2 + x + 2}} dx$	(k) $\int e^{-2x} \sin 2x dx$	(l) $\int e^x \tan^{-1}(e^x) dx$
(m) $\int \sec^3 x \tan x dx$	(n) $\int \frac{1}{1 - \cos 2x} dx$	(o) $\int \frac{1}{1 + e^x} dx$
(p) $\int \frac{x-3}{x^2 - 2x + 4} dx$	(q) $\int (\sec x + 3 \tan x)^2 dx$	(r) $\int w^2 \tan^{-1} w dw$

Source of Question: DHS JC1 Mid-Year CT 9758/2018/01/Q62. (a) Find $\int (\tan^2 x + \cos^2 x) dx$. [4]

(b) Given that $3 - 2x = A(-2 - 2x) + B$ for all values of x , find the constants A and B . Hence find $\int \frac{3 - 2x}{\sqrt{(3 - 2x - x^2)}} dx$. [4]

3. Let $A = \int_0^{\frac{\pi}{2}} x^2 \cos^2 x dx$ and $B = \int_0^{\frac{\pi}{2}} x^2 \sin^2 x dx$.

Find the exact values of

(i) $A + B$ [2](ii) $A - B$ [5]Hence or otherwise, find the exact values of A and B . [2]

4. (i) Show that $\cos^5 x = \cos x - 2 \cos x \sin^2 x + \cos x \sin^4 x$. [2]

Hence find the exact value of $\int_0^{\frac{\pi}{2}} \cos^5 x \, dx$. [3]

(ii) Find also the exact value of $\int_0^{\frac{\pi}{2}} \sin^2 x \cos^5 x \, dx$. [3]

5. Consider the function defined by $f(x) = e^{-x} \sin x$ for all real numbers x .

(a) (i) Write down the first four derivatives of f . [5]

(ii) Hence find a relation between f and its fourth derivative, denoted by $f^{(4)}$. [1]

(iii) Hence write down an anti-derivative of f . [2]

(b) For all nonnegative integers n , let $I_n = \int_{2n\pi}^{(2n+1)\pi} f(x) \, dx$.

(i) Show that for all nonnegative integers n , we have $I_n = \frac{e^{-2n\pi}}{2} (e^{-\pi} + 1)$. [3]

(ii) Hence show that the sequence follows a geometric progression, and deduce the limit of the sequence, explaining clearly your reasoning. [3]

Source of Question: DHS JC2 Mid-Year CT 9758/2018/01/Q5

6. Show that $\frac{d}{dx} \left[(\sqrt{x})^x \right] = \frac{1}{2} (\sqrt{x})^x (1 + \ln x)$. [4]

Hence by using the substitution $u = \sqrt{x}$, evaluate $\int_1^2 (u)^{u^2+1} (1 + 2 \ln u) \, du$ exactly. [4]

Source of Question: HCI JC1 Mid-Year CT 9758/2018/01/Q11

7. (a) By considering $\frac{4x+7}{x^2+2x+5} = \frac{A(2x+2)+B}{x^2+2x+5}$ where A and B are constants, find

$$\int \frac{4x+7}{x^2+2x+5} dx. \quad [3]$$

- (b) Differentiate $\sin(x^2)$. Hence find $\int x^3 \cos(x^2) dx$. [4]

- (c) (i) Given that $t = \tan\left(\frac{x}{2}\right)$ and by considering the double angle formula of

$$\sin x, \text{ show that } \sin x = \frac{2t}{1+t^2}. \quad [2]$$

- (ii) Using the substitution $t = \tan\left(\frac{x}{2}\right)$ and the result in part (i), show that

$$\int \frac{1}{\sin x - 1} dx \text{ can be expressed as } \int \frac{-2}{(t-1)^2} dt \text{ and hence, find}$$

$$\int \frac{1}{\sin x - 1} dx. \quad [4]$$

Source of Question: DHS Prelim 9758/2022/01/Q3

8. (a) Differentiate $e^{\sin^2 2x}$ with respect to x . [2]

(b) Find $\int \frac{e^{\sin^2 2x} \sin 4x}{\sqrt{1+e^{\sin^2 2x}}} dx$. [2]

- (c) Find the exact value of $\int_0^{\frac{\pi}{4}} e^{\sin^2 2x} \sin 4x \cos^2 2x dx$. [3]

Source of Question: RI JC2 Mid-Year CT 9758/2018/01/Q4

9. (i) Find $\int \frac{\cos x}{\sin^{2n} x} dx$, where n is an integer. [1]

- (ii) Use the substitution $x = 2 \tan v$, $0 < v < \frac{\pi}{2}$, to show that

$$\int \frac{1}{x^4 \sqrt{4+x^2}} dx = k \int \frac{\cos v}{\sin^4 v} - \frac{\cos v}{\sin^2 v} dv,$$

where k is a constant to be determined.

Hence, find $\int \frac{1}{x^4 \sqrt{4+x^2}} dx$. [5]

